Chapter 5 exercises. Functions, counting, and cost proofs

DM05-E01 • One formula with different types

For $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = 3n - 7$, determine injectivity, surjectivity, and invertibility, with explicit quantified witnesses. Change only the codomain to make a bijection and give its inverse. Then change both domain and codomain to rationals and decide again.

DM05-E02 • Nonempty fibers and named servers

Send five labeled nodes to three named servers, with every server receiving a node. Use the Stirling recurrence to derive needed values and count assignments. Check using alternating weights for missed servers. Why does the answer differ with unnamed servers?

DM05-E03 • A finite-set equivalence

Prove a self-map of a finite set is injective iff onto, including the empty set. On $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, give self-map counterexamples to each direction for infinite sets.

DM05-E04 • Can composition hide collisions?

For $f : A \to B$ and $g : B \to C$, prove $g \circ f$ injective iff f and $g|_{f(A)}$ are injective. Give explicit sets with injective composition but noninjective g. Give onto composition with f not onto, and identify an additional condition restoring that inference.

DM05-E05 • What pigeonholes guarantee

For 23 integers modulo 22, how many congruent integers are guaranteed? Show three cannot be guaranteed. For positive integers $k, t \geq 1$, with k boxes, find the least total forcing t objects in some box, proving sufficiency and giving a counterexample with one fewer object.

DM05-E06 • Numeric value and bit length

For integer $N \geq 1$, repeatedly replace it by $\lfloor N/2 \rfloor$ until zero. In terms of initial bit length b, find exact iterations. Analyze fixed-word representation, copying the updated remaining bits at every step, and adjusting a read boundary. In the copying model represent zero by an empty string; also state the difference if zero uses one character. Distinguish a fixed word limit from a parameterized asymptotic model and state excluded costs.

DM05-E07 • The full criterion for a quotient formula

For positive m, n, study $h : \mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $h([a]_m) = [a]_n$. Prove well-definedness iff $n \mid m$. In the valid cases give surjectivity and injectivity criteria, and list fibers for $m = 6, n = 3$.

DM05-E08 • Structural counting with partitions

Prove every partition can be the nonempty fibers of a function, with equal-value relation exactly the partition's equivalence relation. For nonnegative integer domain and codomain sizes m, n and image size k, derive the count and the identity summing over all k. Explain empty-set cases using the counting conventions $0^0 = 1$ and $S(0, 0) = 1$. Check $m = 4, n = 3$.

DM05-E09 • Identities and operation tables

For a set of size $n \geq 1$ and specified e, count closed binary operations with two-sided identity e, then those also commutative. Why can requiring some identity be counted by multiplying by n? Evaluate at three. On a set with at least two elements, construct an operation where every element is a left identity but no two-sided identity exists; explain whether this is possible for a singleton.

DM05-E10 • Keep the correct objects in existence proofs

Prove any seven integers have a nonempty contiguous subsequence whose sum is divisible by seven. Why are residues of only the seven original values insufficient? Prove 17 distinct reals contain a strictly increasing or decreasing subsequence of length five, and give a sixteen-term counterexample to that guarantee.

DM05-E11 • Proving binary exponentiation

Prove the lecture's program initialized by $r = 1, b = a, e = N$ correct for every nonnegative integer N, with associative multiplication and multiplicative identity 1, including $N = 0$. Trace 2^{11}, recording e, b, r and updated r. Give the general multiplication count and explain why it does not prove logarithmic bit time for arbitrary integers.

DM05-E12 • Witnesses for growth and averages

Prove $7n^2 + 2n + 1 \in \Theta(n^2)$ with constants, and negate the quantifiers to prove it is not $O(n)$. A length-five linear search has absence probability $1/4$ and conditionally uniform successful positions. Find expected comparisons. If all successes are instead at the first cell, still with success probability $3/4$, is the expectation unchanged?

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 5.

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https://github.com/xinhi6716/computer-science-self-study

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